

PROPOSIÇÃO: Sejam $x, y, z \in \mathbb{R}$. São verdadeiras:

1. $x \leq x$

2. Se $x \leq y$ e $y \leq x$ então $x = y$.

3. Se $x \leq y$ e $y \leq z$ então $x \leq z$.

4. Para $x, y \in \mathbb{R}$, uma e apenas uma dessas possibilidades ocorre:

$$x > y \quad \text{ou} \quad x = y \quad \text{ou} \quad x < y.$$

Intervalos Reais

Sejam $a, b \in \mathbb{R}$; $a < b$. Definimos:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$

↳ intervalo aberto. também denotado por $]a, b[$

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

↳ intervalo fechado

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$

↳ intervalo aberto a esquerda e fechado a direita. Outra notação: $]a, b]$

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

↳ intervalo fechado a esquerda e aberto a direita. Outra notação: $[a, b[$.

$$[a, +\infty) = \{x \in \mathbb{R} : a \leq x\}$$

$$(a, +\infty) = \{x \in \mathbb{R} : a < x\}$$

$$(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$$

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$$

$$(-\infty, +\infty) = \mathbb{R}.$$

} ilimitados

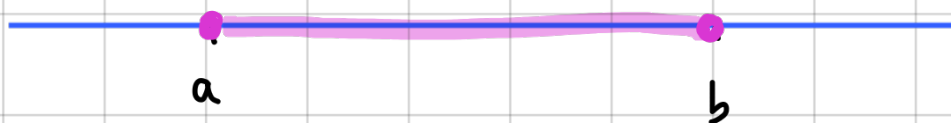
Importante: Um intervalo real (a, b) , com $a < b$, possui uma infinidade de elementos.

Um intervalo é um subconjunto conexo de $\mathbb{R}$. São representados como segmentos da reta real ou semiretas.

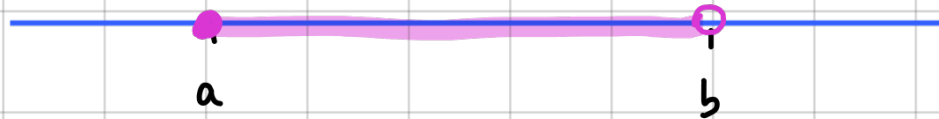
Representação Geométrica

Sejam $a, b \in \mathbb{R}; a < b$.

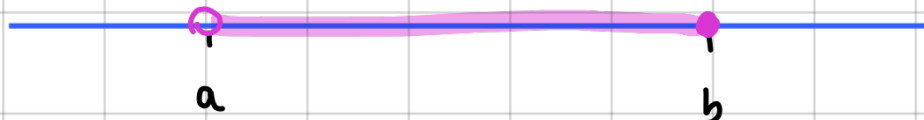
$[a, b]$



$[a, b)$



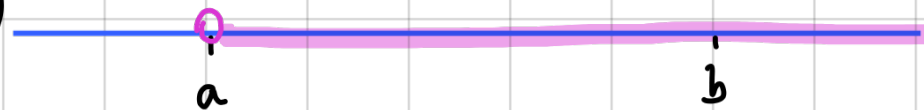
$(a, b]$



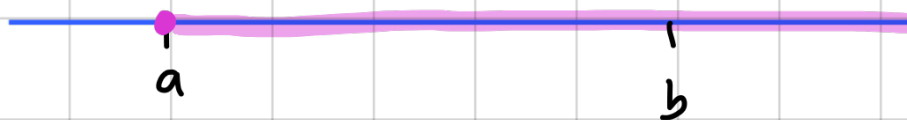
(a, b)



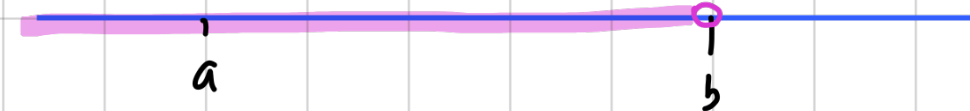
$(a, +\infty)$



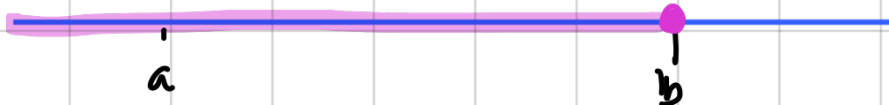
$[a, +\infty)$



$(-\infty, b)$



$(-\infty, b]$



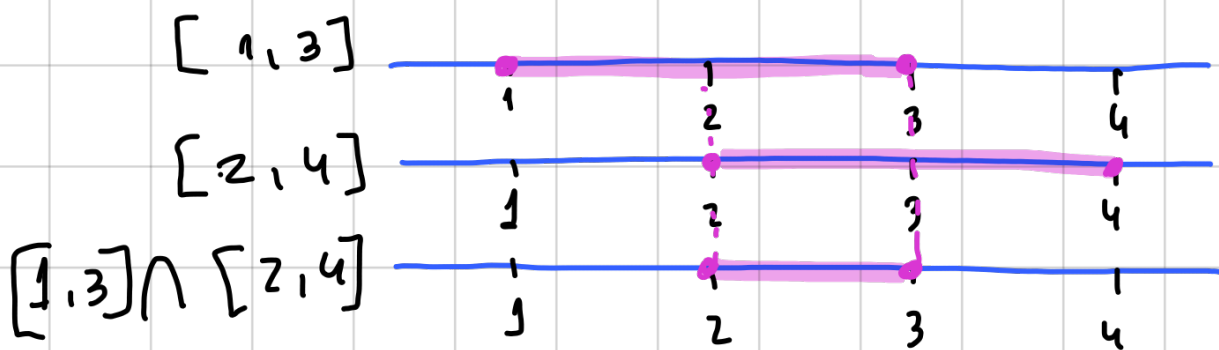
Operações entre intervalos

→ Interseção

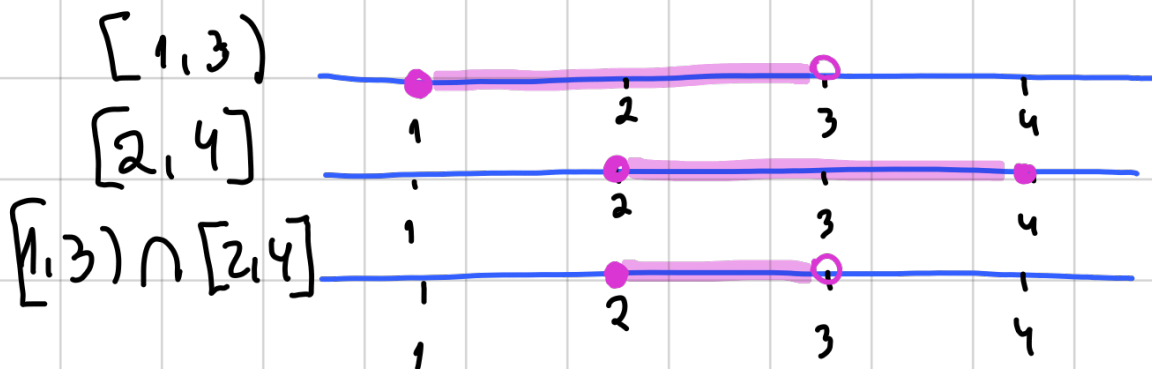
$$[1, 3] \cap [2, 4] =$$

$$= \{ x \in \mathbb{R} : 1 \leq x \leq 3 \text{ e } 2 \leq x \leq 4 \}$$

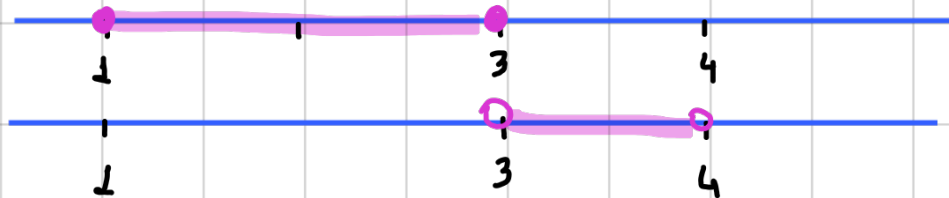
$$= \{ x \in \mathbb{R} : 2 \leq x \leq 4 \}$$



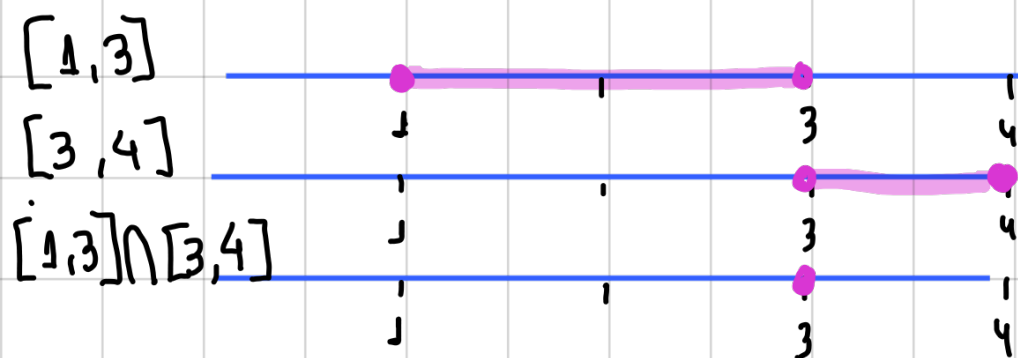
$$[1, 3) \cap [2, 4]$$



$$[1,3] \cap (3,4) = \emptyset$$



$$[1,3] \cap [3,4] = \{3\}$$



Dados dois intervalos I_1 e I_2 , há três possibilidades:

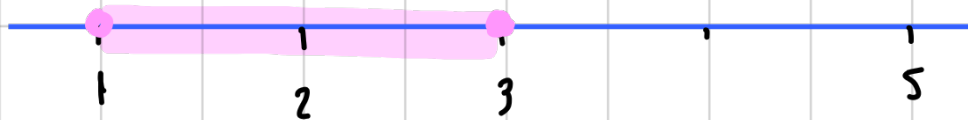
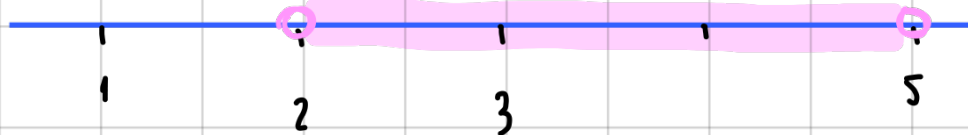
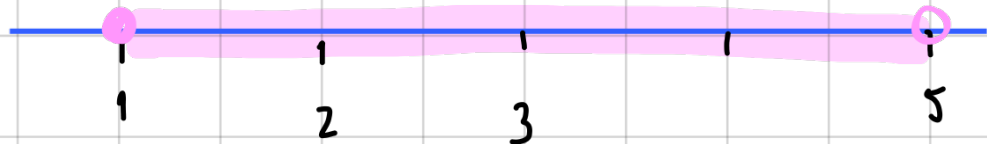
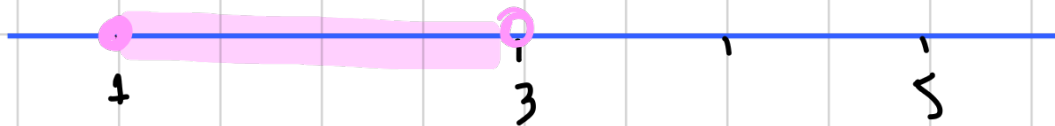
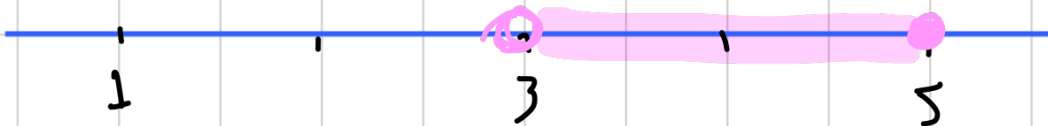
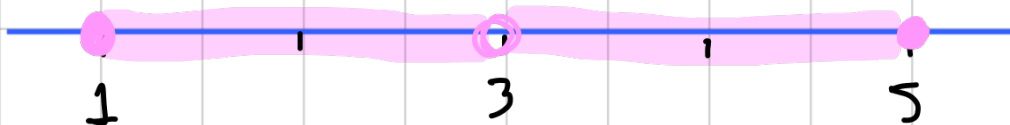
$$I_1 \cap I_2 = I_3 \rightarrow \text{intervalo}$$

$$I_1 \cap I_2 = \emptyset \rightarrow \text{vazio}$$

$$I_1 \cap I_2 = \{a\} \rightarrow \text{apenas um ponto.}$$

→ União

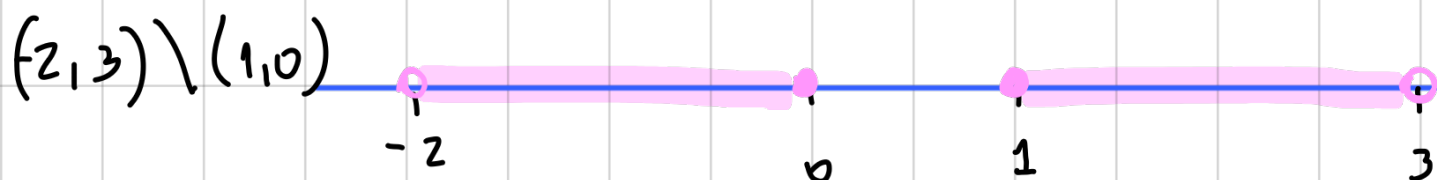
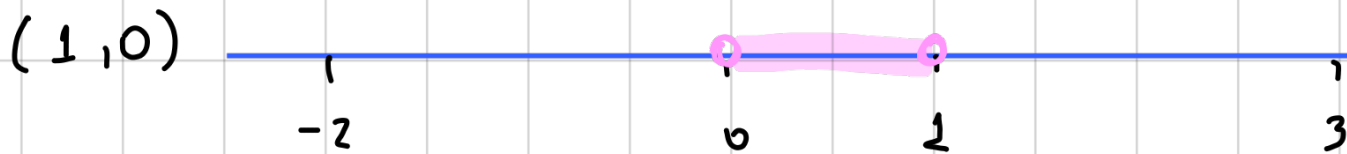
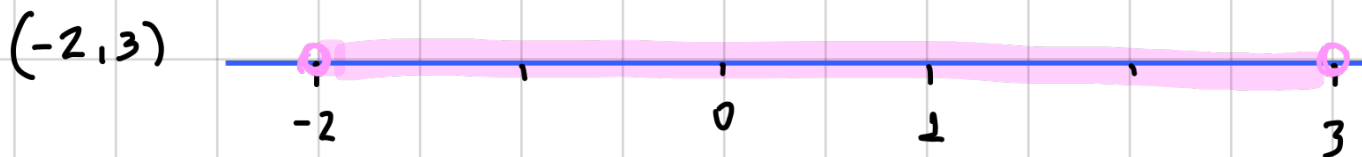
$$[1,3] \cup (2,5)$$

$[1, 3]$  $(2, 5)$  $[1, 3] \cup (2, 5)$  $= [1, 5)$ $[1, 3) \cup (3, 5]$ $[1, 3)$  $(3, 5]$  $[1, 3] \cup (3, 5]$ 

↳ nesse caso, a união de dois intervalos não resultou em um novo intervalo.

Diferença entre Intervalos

$$(-2, 3) \setminus (1, 0)$$

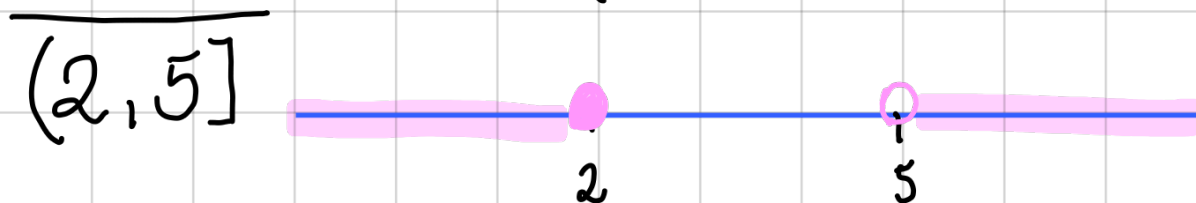
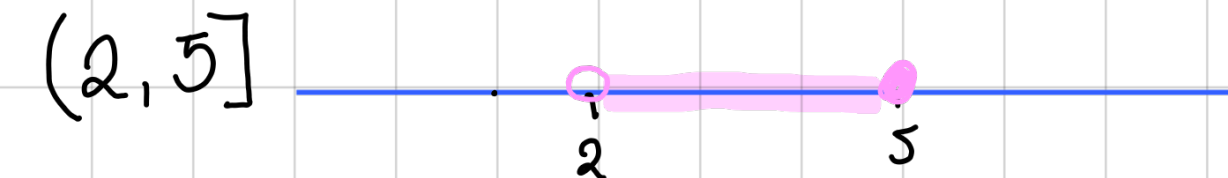


$$= (-2, 0] \cup [1, 3).$$

Complementar de um Intervalo

O complementar de um intervalo real será seu complementar em relação a $\mathbb{R}$ (salvo quando dito o contrário).

Exemplo



$$\hookrightarrow (-\infty, 2] \cup (5, +\infty).$$